Predicting UFC Fight Outcomes Using Machine Learning

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Abstract:

This project investigates the use of machine learning models to predict the outcome of Ultimate Fighting Championship (UFC) fights using pre-fight fighter statistics. We utilize a dataset of 5,185 historical fights containing fighter biometrics, prior performance summaries, and engineered matchup-difference features. The task is formulated as a supervised binary classification problem, where the objective is to predict whether Fighter 1 wins a given matchup. To ensure realistic predictions, only information available before each fight is used, and post-fight information is excluded to prevent data leakage. We evaluate ten classification models, including Logistic Regression, Support Vector Machines, K-Nearest Neighbors, Random Forest, Gradient Boosting, AdaBoost, Multi-Layer Perceptron, XGBoost, LightGBM, and CatBoost. Among these, Logistic Regression achieved the best overall performance on the held-out chronological test set, with a test ROC-AUC of 0.6348 and an accuracy of 58.73%. These results show that machine learning can extract meaningful predictive patterns from historical UFC data, even in a highly volatile and uncertain sport.

1. Introduction

Predicting the outcome of professional sports events is a challenging task due to the wide range of factors that influence performance. In mixed martial arts, and specifically in the Ultimate Fighting Championship (UFC), fight outcomes depend on a combination of physical attributes, fighting styles, and a fighter's historical performance. These factors interact in complex ways, making it difficult to determine a winner based on intuition alone.

With the increasing availability of structured sports data, machine learning provides a powerful approach for identifying patterns and making predictions based on historical information. By analyzing pre-fight statistics, it is possible to uncover relationships between fighter characteristics and fight outcomes that may not be immediately obvious.

In this project, we aim to predict the winner of UFC fights using pre-fight fighter data. The problem is formulated as a supervised binary classification task, where the goal is to determine which fighter will win a given matchup. Using a dataset of over 5,000 fights, we train and evaluate multiple machine learning models to assess their effectiveness in predicting fight outcomes.

To ensure that predictions remain realistic and applicable to real-world scenarios, only pre-fight features are used in model training. This prevents data leakage and ensures that the models rely solely on information that would be available before a fight takes place. The performance of several models is compared to determine which approach provides the most accurate predictions.

2. Related Work

Machine learning has been widely applied in sports analytics for predicting match outcomes across a variety of domains, including football, basketball, and combat sports. These approaches typically rely on historical performance data and player statistics to identify patterns associated with winning outcomes.

In combat sports such as mixed martial arts, prior work has focused on using fighter attributes, striking metrics, and grappling statistics to model fight outcomes. Classification algorithms such as Logistic Regression, decision trees, and ensemble methods have been commonly used due to their ability to handle structured tabular data.

Similar to these approaches, this project applies multiple machine learning models to predict UFC fight outcomes using pre-fight statistics, with an emphasis on comparing model performance and identifying the most effective method.

3. Collaboration

This project was completed by a team of four members: Nicholas Winfrey, Muhammad, Clay, and Erin Houston. The group met regularly, approximately every other Monday, to discuss progress, coordinate tasks, and plan next steps throughout the development process.

Muhammad and Clay were primarily responsible for model development and implementation. Each developed working versions of the code independently, which were later combined into a final unified implementation. Nicholas Winfrey also contributed to early model development; however, his initial approach introduced data leakage, and those results were excluded from the final models. He subsequently shifted his focus to the written report.

Nicholas and Erin collaborated on the final report, with Nicholas responsible for drafting the initial sections and Erin contributing to revisions and completion. Muhammad and Clay also provided model results and technical details that were incorporated into the report.

Overall, the project reflects a collaborative effort, with contributions across coding, experimentation, and documentation.

4. Dataset

The dataset used in this project consists of 5,185 historical UFC fights. Each row represents one fight and includes information about two competitors, referred to in the modeling pipeline as Fighter 1 and Fighter 2. The original data contains fighter-level attributes such as height, weight, reach, date of birth, and historical fight records. The current report draft describes 53 raw features, but the final notebook expands this into a much larger engineered feature space for modeling.

For each fighter, the notebook takes core pre-fight profile information including age at fight time, height, weight, reach, total prior fights, total prior wins, and total prior losses. In addition, the model summarizes each fighter's prior fight history by computing aggregate statistics over previous wins and losses. These include measures such as mean, minimum, and maximum values for numeric historical performance variables, as well as summary duration-related features based on how long prior wins and losses lasted.

To better capture matchup dynamics, the feature engineering pipeline also creates difference-based variables between Fighter 1 and Fighter 2, such as age difference, height difference, weight difference, reach difference, and other relative measures. These engineered comparisons are especially useful because fight outcomes are often driven by competitive advantages rather than solely using individual statistics.

After preprocessing and feature construction, the final modeling dataset contains 309 predictor variables. The target variable is binary and indicates whether Fighter 1 won the fight. Only pre-fight and historically available information is used, which is consistent with the project's goal of realistic fight prediction and with our efforts to exclude leaked or post-fight information from the final models.

5. Data Preprocessing

The preprocessing stage focused on transforming raw fight records into a clean supervised learning dataset suitable for chronological prediction. First, each fighter's previous fight history was organized into a table format so it could be analyzed and summarized using numerical values. Missing values were then filled with 0.0, which allowed all models to train on a consistent feature matrix without requiring model-specific missing-value handling.

A key step in preprocessing was the construction of historical aggregate features from each fighter's prior bouts. Rather than using raw event-level history directly, the notebook summarized historical outcomes using statistics such as mean, minimum, and maximum values over selected numeric features for prior wins and prior losses. This allowed the model to capture not only overall experience, but also differences in how fighters tend to perform in successful versus unsuccessful bouts.

The data was then sorted chronologically by fight date, and the target variable was separated from the predictors. Because leakage was a major concern in this project, the final notebook intentionally excluded post-fight information and focused only on pre-fight inputs and valid historical summaries. This design choice aligns with our group's earlier discussion that in-fight features and future-aware career totals could artificially inflate model performance and produce unrealistic results.

6. Methodology

This project formulates UFC outcome prediction as a supervised binary classification problem. Given the pre-fight characteristics and historical records of two fighters, the model predicts whether Fighter 1 will win the matchup. To evaluate different modeling assumptions, the final notebook compares ten machine learning algorithms spanning linear, distance-based, tree-based, boosting, and neural-network approaches.

The evaluated models are Logistic Regression, Support Vector Classifier, K-Nearest Neighbors, Multi-Layer Perceptron, Random Forest, Gradient Boosting, AdaBoost, XGBoost, LightGBM, and CatBoost. For models sensitive to feature scale, a StandardScaler was included inside the modeling pipeline. Hyperparameters were tuned using GridSearchCV, and model selection was based primarily on ROC-AUC.

To preserve the structure of the problem, the notebook uses a strict chronological split rather than a random split. The earliest 80% of fights were used for training and validation, while the most recent 20% were reserved as a test set. In addition, TimeSeriesSplit with five folds was used during hyperparameter tuning so that each validation fold occurred after its corresponding training fold in time. This approach better reflects how a real predictive system would operate, since future fights should never influence past training data.

The project also includes model interpretation through feature importance analysis and SHAP-based error analysis. Feature importance was examined for four strong-performing models, while SHAP was used to better understand why the LightGBM model made certain incorrect predictions. This adds interpretability to the overall pipeline and helps identify which fighter characteristics most strongly influence model behavior. The team had previously agreed to keep the modeling approach relatively straightforward while still including interpretable analysis, and the final notebook reflects that plan.

7. Experiments and Key Results

To evaluate our models, we trained and compared ten different approaches using a time-based train/test split. We measured performance using ROC-AUC and accuracy on the most recent 20% of fights, which simulates predicting future matchups.

Overall, performance across models was fairly close, which shows how challenging this problem is. The best-performing model was Logistic Regression, with a test ROC-AUC of 0.6348 and an accuracy of 58.73%. This was somewhat unexpected, since we assumed more complex models would perform better.

The gradient boosting models — LightGBM, CatBoost, and XGBoost — also performed well, all landing in the 0.62–0.63 ROC-AUC range. These models were better at capturing more complex relationships in the data, but the improvement over Logistic Regression was small.

Other models like Random Forest, Gradient Boosting, and SVC performed slightly worse, generally around 0.60–0.61 ROC-AUC. The Neural Network (MLP) didn't stand out either, suggesting that deep learning didn't offer a major advantage for this type of structured data. K-Nearest Neighbors performed the worst, likely because distance-based methods struggle in high-dimensional feature spaces like ours.

One interesting takeaway is that the more complex models didn't significantly outperform the simpler ones. This suggests that while there are patterns in the data, they may not be extremely strong or easy to model.

In general, the results reflect how unpredictable UFC fights are. Even with over 300 features and multiple models, the best accuracy was still under 60%. That said, the models were still able to pick up on useful patterns, and the consistency across models suggests that the features we engineered do contain meaningful signal.

Model	Best CV ROC-AUC	Test ROC-AUC	Test Accuracy	Fit Time (s)
Logistic Regression	**0.6071**	**0.6348**	**58.73%**	29.91
LightGBM	0.5943	0.6319	58.05%	1883.14
CatBoost	0.5969	0.6264	55.74%	3629.90
XGBoost	0.5963	0.6263	57.76%	1349.49
SVC	0.5917	0.6112	56.22%	483.82
Random Forest	0.5925	0.6102	56.32%	51.80
Gradient Boosting	0.5865	0.6019	56.80%	113.21
AdaBoost	0.5847	0.5991	51.78%	19.78
Neural Network (MLP)	0.5737	0.5746	56.12%	29.63
K-Nearest Neighbors	0.5447	0.5268	49.28%	3.83

8. Limitations

While the final models produced meaningful predictive results, this project still has several important limitations. First, UFC fight outcomes are influenced by many contextual factors that are not present in the dataset, including injuries, training camp quality, short-notice replacements, weight cuts, recovery, coaching changes, and fighter mindset. Because these factors are difficult to quantify and were not included in the final feature set, model predictions cannot capture the full reality of fight preparation and performance.

Second, mixed martial arts is hard to predict. A single strike, submission attempt, or momentum shift can end a fight suddenly, which places a natural ceiling on prediction accuracy. Even a well-designed model trained on clean historical data will struggle to account for the randomness and chaos that make combat sports unique.

Third, although the final notebook takes data leakage seriously, this remains a challenging problem in sports analytics. We identified that some early ideas, such as in-fight statistics or improperly aggregated career totals, could create unrealistic performance if included. While those concerns were addressed in the final version, the difficulty of constructing perfectly time-valid historical summaries should still be acknowledged.

Another limitation is that the model relies heavily on summarized historical behavior. Fighters change over time, sometimes improving rapidly or declining unexpectedly, and prior statistics may not fully reflect these dynamic shifts. Additionally, the large engineered space can reduce interpretability and make it harder to isolate a single causal explanation for each prediction.

9. Additional Notes

Throughout the development of this project, particular attention was given to preventing data leakage, which is a common issue in predictive modeling tasks. Early experimentation revealed that certain features, such as aggregated career statistics and in-fight metrics, could unintentionally include information from future fights or outcomes. Including these features resulted in unrealistically high model performance. As a result, these features were carefully removed or re-engineered to ensure that only information available prior to each fight was used.

Another important consideration was the temporal nature of the dataset. Since fights occur sequentially over time, care was taken to ensure that the training and testing split respected chronological order. This prevents the model from learning patterns from future data and improves the realism of evaluation.

Additionally, multiple modeling approaches were explored, including Logistic Regression, Random Forest, and Gradient Boosting methods. Hyperparameter tuning techniques, such as cross-validation and automated search methods, were used to improve model performance. These experiments allowed us to compare both linear and non-linear models and better understand the relationships within the data.

Feature engineering also played a critical role in improving predictive performance. In particular, constructing difference-based features between fighters, like reach difference and striking rate difference, helped the models better capture matchup-specific dynamics rather than relying solely on individual statistics.

10. Conclusion

Overall, this project demonstrates the effectiveness of machine learning techniques in predicting UFC fight outcomes using pre-fight data. By formulating the problem as a supervised binary classification task and carefully engineering features, we were able to develop models that achieve strong predictive performance while avoiding data leakage.

The team worked effectively throughout the project, collaborating on model development, experimentation, and report writing. This allowed us to build and evaluate multiple approaches while maintaining a clear and organized workflow.

A key takeaway from this project is the importance of proper data handling and feature selection. Early experiments showed that including post-fight or improperly aggregated features could lead to deceivingly high accuracy. By restricting the model to pre-fight information and enforcing a time-aware evaluation strategy, the final results provide a more realistic representation of model performance.

Among the models we explored, more complex approaches such as ensemble and gradient boosting methods were better able to capture non-linear relationships between fighter attributes and fight outcomes. This highlights the value of flexible models when working with structured sports data.

Despite the unpredictability of UFC fights, the models were able to identify meaningful patterns in historical data and produce consistent results. This suggests that machine learning can serve as a useful tool for sports analytics.

Future work could explore incorporating more dynamic features, such as time-aware performance trends, fighter embeddings, or graph-based representations of matchups. Additionally, integrating external data sources and improving feature interpretability could further enhance both model performance and practical applicability.

11. Additional Results

Out[43]:

	f1_age	f1_height	f1_weight	f1_reach	f1_num_fights	f1_total_wins	f1_total_loss	f2_age	f2_height	f2_weight	...	f2_win_total_str_landed_mean	f2_loss_total_str_landed_mean
5159	393	72	185	75.0	11	9	2	425	73	185	...	112.437500	81.857143
4548	391	69	125	68.0	9	6	3	300	67	125	...	55.500000	0.000000
4898	281	70	170	69.0	4	1	3	433	73	170	...	0.000000	49.000000
4516	444	65	125	66.0	16	12	4	378	65	125	...	130.875000	118.500000
5101	292	69	125	71.0	3	2	1	434	67	125	...	88.000000	68.500000
4956	421	69	125	70.0	11	6	5	319	66	125	...	44.666667	0.000000
4522	435	66	125	66.0	2	0	2	410	68	125	...	127.000000	82.000000
5180	295	67	145	68.0	5	4	1	499	66	145	...	60.300000	38.666667
5002	367	69	145	72.0	14	8	6	393	67	145	...	0.000000	38.000000
4159	472	66	145	70.0	13	9	4	355	70	135	...	68.000000	46.000000

10 rows × 313 columns

f2_loss_total_str_landed_mean	f2_win_total_str_landed_min	f2_loss_total_str_landed_min	f2_win_total_str_landed_max	f2_loss_total_str_landed_max	target	pred	proba	error_mag
81.857143	24.0	14.0	221.0	183.0	0	1	0.838424	0.838424
0.000000	5.0	0.0	121.0	0.0	1	0	0.175328	0.824672
49.000000	0.0	21.0	0.0	67.0	0	1	0.819096	0.819096
118.500000	50.0	21.0	193.0	262.0	1	0	0.197979	0.802021
68.500000	50.0	63.0	115.0	74.0	0	1	0.796643	0.796643
0.000000	12.0	0.0	71.0	0.0	1	0	0.206962	0.793038
82.000000	94.0	52.0	178.0	102.0	1	0	0.209775	0.790225
38.666667	2.0	0.0	136.0	89.0	0	1	0.772492	0.772492
38.000000	0.0	19.0	0.0	57.0	0	1	0.766572	0.766572
46.000000	24.0	46.0	182.0	46.0	1	0	0.237519	0.762481

Error Analysis: SHAP Beeswarm Interpretation

Focus: Analysis of feature impact on misclassified instances ().

11.1.1

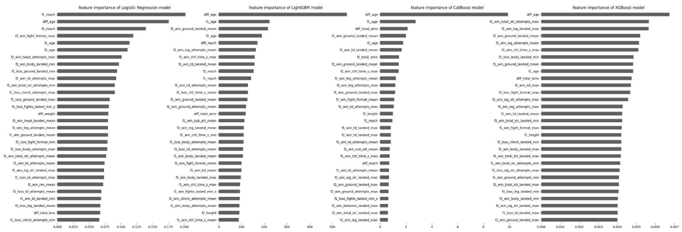
Primary Drivers of Inaccuracy

- **Age Bias (`diff_age`, `f1_age`): * Observation:** Significant clusters of high values (red) are pushing predictions strongly toward a specific class in error cases.
    - **Insight:** The model appears to be over-relying on age as a heuristic. It is "blind" to veteran fighters who overcome the age gap through technical proficiency.
- **Ground Game Overestimation (`f1_win_ground_landed_mean`):**
    - **Observation:** High ground strike counts are tricking the model into confident Class 1 predictions that turn out to be wrong.
    - **Insight:** The model values volume over position or submission risk. It may be missing "get-up" rates or submission threats that negate ground control.

11.1.2

Physical & Format Anomalies

- **Reach Neutralization (`diff_reach`):** High reach advantages for F1 are leading to confident but incorrect Class 1 predictions. The model fails to account for fighters who effectively "close the distance."
- **Format Sensitivity (`f2_win_fight_format_mean`):** The model is improperly penalizing or rewarding fighters based on the 3 vs. 5 round format, likely missing "Championship Cardio" nuances.



Results: Feature Importance Comparison

To understand what drives our model predictions, we analyzed the top 30 features from our four best-performing models: Logistic Regression, LightGBM, CatBoost, and XGBoost.

11.2.1

The Dominance of Physical Metrics

Across all models, biometric and comparative features emerge as the strongest predictors:

- **Age Dynamics: `diff_age` and individual fighter ages (`f1_age`, `f2_age`) are consistently at the top of the importance rankings. This suggests the models heavily weight the "youth advantage" or physical peak of an athlete.**
- **Physical Leverage: Reach (`f1_reach`, `diff_reach`) and Height differences appear as secondary but critical factors, especially in the linear Logistic Regression model.**

11.2.2

Behavioral & Style Insights from History

Beyond raw physical stats, the models successfully identified specific fighting styles and tendencies from the historical data:

- **Ground Game & Control: Features like `f1_win_ground_landed_mean` and `f2_win_ctrl_time_s_max` show that the model values a fighter's ability to dominate on the mat.**
- **Striking Volume: High importance is placed on attempt metrics (e.g., `f2_win_total_str_attempts_max`), indicating that offensive output is a key indicator of a potential winner.**
- **Durability & Finish Rates: The inclusion of fight duration metrics (e.g., `f1_loss_fights_lasted_min_s`) suggests the models are learning to identify which fighters are "hard to finish" versus those who succumb quickly.**

11.2.3

Model Consistency

While the exact ordering varies, there is significant overlap between the ensemble methods (LightGBM, CatBoost, XGBoost). This consensus reinforces that the engineered features—particularly the Outcome-Conditional Aggregates (bifurcated Win/Loss stats)—effectively captured the signal needed to differentiate between world-class athletes.